Towards Faithful LLM-Based Narrative Summarization of Individual GPS Trajectories

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Abstract

Enriching raw GPS trajectories with semantic context (points of interest (POIs), activities, transportation modes) has been shown to improve downstream mobility tasks such as trajectory classification and next-POI recommendation, but progress on these tasks remains limited by the scarcity of ground-truth trajectory data. This paper investigates an alternative angle: converting an individual's enriched GPS trajectory into a natural-language narrative using Large Language Models (LLMs), so that consolidated text classification, retrieval, and analysis methods can be applied to mobility data instead of bespoke spatio-temporal models. We present a new pipeline and an empirical study on the GeoLife dataset. Applying it in practice surfaces challenges not anticipated from a purely methodological standpoint: LLMs default to plausible-sounding but unsupported transportation-mode claims, naive POIs selection introduces narratively misleading anchors, and narrative coverage of the original spatial facts collapses sharply as trajectory complexity grows. Standard reference-based text-generation metrics (ROUGE, BERTScore) cannot detect any of this, since they require gold summaries that do not exist for this task; we discuss the reference-free, structure-grounded evaluation approach we designed instead. We argue these findings are relevant beyond this specific pipeline: applying consolidated information-retrieval and natural-language-generation methodology to complex, geographically grounded data opens evaluation and faithfulness challenges that classic text-generation benchmarks do not cover.

Keywords

Semantic trajectories, Large Language Models, Trajectory summarization, Human mobility

1. Introduction

Raw GPS trajectories – sequences of timestamped coordinates – are, on their own, largely uninformative: they say where a device was, not what its owner was doing or why. *Semantic trajectories* address this by enriching the raw signal with contextual layers such as stops and moves, points of interest, activity types, and transportation modes [3, 2]. This added context has repeatedly been shown to improve downstream mobility tasks: transportation-mode and trajectory classification benefit from semantically or textually informed representations [9, 6], and next-POI recommendation improves when trajectory history is combined with richer contextual or knowledge-graph signal [11].

Progress on these tasks is nonetheless bottlenecked by data availability. Large-scale, richly annotated trajectory datasets are scarce: GPS traces are personally identifying, collecting them at scale raises serious privacy concerns, and the datasets that do exist at scale are frequently proprietary (held by telecom operators, ride-hailing platforms, or mapping companies) and unavailable for open research or for building the kind of ground-truth-labeled benchmarks that would let classification and recommendation methods be trained and evaluated rigorously [4].

This paper investigates a different route toward the same downstream goal. Rather than engineering task-specific numeric or graph-based models directly over structured trajectory data, we ask whether *converting* a trajectory – a sequence of enriched spatio-temporal points – into a natural-language textual description via an LLM can let us reuse consolidated, mature text classification and retrieval methodology for tasks like trajectory classification and next-POI prediction. Prompting LLMs with textual trajectory representations is already an established paradigm for next-location prediction [7, 8];

Proceedings of the 16th Italian Information Retrieval Workshop, IIR 2026, November 12–14, 2026, Rome, Italy
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a symmetric body of work goes in the opposite direction, using LLMs to *generate* plausible synthetic trajectories from textual descriptions or personas [10], and question-answering benchmarks have begun probing whether LLMs can semantically *understand* a given trajectory when asked about it [5]. What remains largely unexplored is the middle step: producing a faithful, free-form, human-readable narrative *of one individual's actual trajectory*, as an intermediate representation in its own right rather than as an answer to a fixed question or as training material for a downstream generator.

We propose a methodology to do exactly this, extending our own prior work on aggregate, place-level natural-language description from geolocated data [1] down to the level of a single moving individual. At a high level, the pipeline (i) segments a raw GPS trajectory into alternating stop and move episodes, (ii) semantically enriches stop episodes with nearby POI and move episodes with distance, duration, and transportation mode, and (iii) prompts an instruction-tuned LLM, under several prompting strategies, to narrate the resulting episode sequence as natural-language text.

Carrying out a first full-scale implementation of this idea, however, surfaced problems that a purely top-down design would not have anticipated, and that motivate this paper as much as the pipeline itself: LLMs asked to narrate mobility data introduce specific, quantifiable forms of unfaithfulness – defaulting to a plausible transportation mode when none is given, treating an incidentally nearby but irrelevant POI as if it were a meaningful destination, and progressively dropping concrete place references in favor of vague language as the underlying trajectory grows more complex. None of these failure modes is visible to standard natural-language-generation metrics such as ROUGE or BERTScore, which additionally require reference summaries that, to the best of our knowledge, do not yet exist at any scale. From an information-retrieval standpoint, this is the point worth stressing: applying consolidated IR methodology – text generation, established similarity metrics, text-based downstream classification – to a data type as structurally rich and factually constrained as a semantic trajectory is not a routine transfer.

2. Methodology

Our pipeline reframes the four-step, aggregate-level design of [1] at the individual-trajectory level, operating on one user's raw GPS stream at a time.

Segmentation. Each trajectory is split into alternating *stop* and *move* episodes using classic distance- and time-threshold stay-point detection (a point is part of a stay if enough subsequent points remain within a fixed radius for at least a minimum dwell time; everything else is a move). The detector processes each recorded trajectory file as a single continuous sequence: GPS signal gaps (e.g., from underground travel) are neither split into separate trajectories nor bridged by any inference – whatever the detector naturally produces across a gap is accepted as-is, since bridging unobserved time would itself be a faithfulness violation.

Semantic enrichment. Each stop episode is enriched with the nearest point of interest from an OpenStreetMap[1] extract (name and category, e.g.,"bank", "restaurant", within a fixed search radius). Each move episode is enriched with its distance, duration, and average speed, and, where the dataset provides a genuine ground-truth transportation-mode label for that time interval, with that label. Labels are attached exclusively via direct, interval-level ground-truth lookup and never inferred from speed or any other heuristic; a move with no matching label is left unlabeled. This is a deliberate methodological constraint: the narration step (below) explicitly forbids the LLM from guessing an unstated mode, and the enrichment step must not launder a guess into the input as if it were a fact.

Merging. A lightweight cleanup pass collapses spurious (stop, tiny move, stop) triples caused by GPS jitter around the stay-detection threshold back into a single stop, re-enriching the merged location. This never bridges unobserved time – only continuously observed points are involved.

Narration. The resulting ordered episode sequence is formatted as a numbered list of events (time range, duration, and either a destination/POI for stops or distance and mode-if-known for moves) and given to an instruction-tuned LLM under several distinct prompt templates – a neutral third-person

[1] https://wiki.openstreetmap.org/wiki/Map_features

summary, a first-person diary entry, and a length-constrained concise variant – mirroring the multi-prompt-per-input methodology of [1]. Every template includes explicit anti-hallucination instructions: use only the facts given, do not invent places, activities, or durations, and refer to an unlabeled move neutrally rather than guessing a mode.

3. Experiments

We ran the segmentation and enrichment steps over the entire GeoLife[2] dataset (182 users, 18,670 recorded trajectories, approximately 25 million GPS points). Narrating every trajectory with an LLM was computationally infeasible at this scale with the resources available (Qwen2.5-1.5B-Instruct [12]).

Evaluating the resulting narratives required a design decision of its own. ROUGE and BERTScore, the metrics used for the aggregate-level task in [1], are reference-based: they require a human-written gold summary to compare against, and no such gold summaries exist yet for individual-trajectory narration at any useful scale. We therefore designed two reference-free checks that instead compare a generated narrative directly against the structured facts it was given: a *mode-hallucination check*, flagging any stated transportation mode not backed by a ground-truth label on the corresponding move episode, and a *grounding check*, which for every episode determines whether it is mentioned in the text at all (*coverage*), whether a stated duration is numerically consistent with the true value, and whether mentioned episodes preserve the trajectory's true chronological order.

Table 1
Aggregate results over 194 generated narrations.

Metric	Value
Mode-hallucination violations	36/194 (18.6%)
Episode coverage	430/952 (45.2%)
Duration accuracy (where stated)	121/180 (67.2%)
Chronological order fidelity	192/239 (80.3%)

Table 1 summarizes the results across the 194 completed narrations. Beyond the aggregate numbers, three findings stand out as open problems for this line of work, each pointing to a different way that a currently mature, well-evaluated technology (instruction-tuned text generation) behaves unpredictably once the underlying data is a geographically and temporally constrained trajectory rather than free text.

Transportation-mode hallucination. 26 of the 36 flagged violations involve the model stating that the person "walked" a given move, even when that move carries no ground-truth mode label whatsoever. Despite an explicit prompt instruction to refer to unlabeled movement neutrally, the model's default prior – walking being the statistically most common everyday mode – overrides the instruction at a non-trivial rate. This is a specific, quantifiable, and reproducible failure of instruction-following under a strong world prior, distinct from open-ended hallucination.

POI relevance filtering is needed. Enriching a stop with its single nearest OpenStreetMap feature occasionally selects street furniture – a surveillance camera, a bench – rather than a meaningful destination, simply because it happens to be the closest tagged object. Handed such a feature as if it were a place the person visited, the LLM narrated it as a fabricated safety concern ("a location that required me to be very careful... stay alert and cautious"), inventing a plausible-sounding but entirely unsupported interpretation from a low-information enrichment artifact. Naive nearest-neighbor POI selection is not enough; a relevance-filtering step over candidate POIs – itself a classic information-retrieval problem, here applied to a geographic candidate set instead of a document collection – is needed before enrichment output reaches the LLM.

The spatial component is the first casualty of compression. Coverage is strongly, negatively correlated with the number of episodes in a trajectory (Pearson's $r = -0.73$, on a -1 to 1 scale):

[2] https://www.microsoft.com/en-us/download/details.aspx?id=52367

narratives of simple trajectories ($\leq$ 3 episodes) retain 92% of the underlying facts on average, while narratives of complex trajectories ($\geq$ 8 episodes) retain only 27%. In the most extreme case, a 30-episode, roughly ten-hour trajectory is compressed into a summary that never names a single actual place, replacing every concrete stop with generic language ("stopping frequently for various reasons such as eating, using the restroom, and shopping"). Under length pressure, the model does not fail loudly with a false claim; it fails quietly by discarding exactly the spatial specificity that would make the narrative useful as a representation for downstream geographic tasks.

Finally, building the evaluation itself was not a solved problem going in. Structural grounding checks of this kind are sensitive to seemingly minor design choices (e.g., how episodes that share a textual anchor are disambiguated from one another, or how a stated duration is attributed to the correct episode rather than a neighboring one). Therefore, designing a metric that properly measures faithfulness is itself a non-trivial, open problem.

4. Conclusion and Future Work

We presented a methodology and a full-scale empirical study for converting individual GPS trajectories into natural-language narratives using LLMs, motivated by the goal of using text as a common, tool-compatible representation for mobility data in place of bespoke structured models – directly useful for tasks such as trajectory classification and next-POI recommendation that are otherwise held back by the scarcity of large, richly labeled, non-proprietary trajectory datasets. The pipeline itself runs end-to-end at scale. The open problem this work surfaces is that the resulting narratives are not faithful for free. Across 194 generated narratives we quantified three distinct, recurring failure modes – default transportation-mode hallucination, POI-relevance noise from naive nearest-neighbor enrichment, and a sharp, measurable loss of spatial specificity as trajectory complexity grows – none of which is visible to standard reference-based text-generation metrics, which in any case require gold summaries that do not exist for this task.

Future work follows directly from these findings. On the modeling side: filtering candidate POIs for narrative relevance before enrichment, strengthening prompts with explicit negative instructions against default-mode guessing, and evaluating larger or API-hosted LLMs against the small open model used here. On the evaluation side: collecting volunteer-authored gold summaries to finally enable ROUGE/BERTScore-style comparison alongside our reference-free grounding check, and extending faithfulness checking beyond keyword- and rule-based matching (which cannot catch fabricated narrative content such as invented social detail or spurious concluding statements) toward an LLM-as-judge protocol. Most directly connected to our original motivation, we plan to close the loop with the downstream tasks that justify this whole direction: measuring transportation-mode classification accuracy and next-POI prediction performance when the input is the generated narrative text itself, rather than the raw or structured trajectory, to quantify how much task-relevant signal a natural-language representation of a trajectory actually preserves.

GenAI usage disclosure

During the preparation of this work, the author(s) used Claude (Antropic) to edit text and code. After using this tool, the author(s) reviewed, edited, and verified the content as needed and take(s) full responsibility for the final publication.

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Version 1.07-ntp
2026-04-09

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